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MILITARY SPECIFICATION

POWER SUPPLY, HIGH VOLTAGE

(FOR TORPEDO MARK 37 MODS 2 AND 3)

This specification has been approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification describes requirements for the procurement of a high voltage power supply used in the Torpedo Mark 37, Mods 2 and 3.

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-387	Plastic-Sheet, Laminated, Thermosetting (for Designation Plates)
FF-N-836	Nuts, Hexagonal and Square
FF-S-85	Screw, Cap, Slotted and Hexagonal Head
FF-S-86	Screw, Cap, Socket Head
FF-S-92	Screw, Machine, Slotted or Cross-Recessed
QQ-A-250/3	Aluminum Alloy Alclad 2014, Plate and Sheet
QQ-A-250/8	Aluminum Alloy 52S; Plate and Sheet
QQ-A-250/11	Aluminum Alloy 61S; Plate and Sheet
QQ-A-367	Aluminum Alloy Forgings, Heat-Treated
QQ-B-613	Brass, Leaded and Non-Leaded; Flat Products (Plate, Bar, Sheet and Strip)
QQ-B-626	Brass, Leaded and Non-Leaded; Rods, Shapes, Forgings and Flat Products with Finished Edges (Bar, Flat Wire and Strip)
QQ-B-637	Brass, Naval, Wire, Shapes and Forgings and Flat Products with Finished Edges
QQ-B-750	Bronze, Phosphor, Bars, Plates, Rods, Sheets, Strips, Flat Wire and Structural and Special Shaped Sections
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-C-390	Copper Alloy Castings (Including Cast Bar)
QQ-C-450	Copper-Aluminum Alloy (Aluminum Bronze) Plate, Sheet, Strip, and Bar (Copper Alloy Numbers 606, 612, 613, 614 and 628)
QQ-N-290	Nickel Plating (Electrodeposited)

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QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-365	Silver Plating (Electrodeposited)
QQ-S-630	Steel, Bar, Carbon, Hot Rolled (Merchant Quality)
QQ-S-634	Steel, Bar, Carbon, Cold Finished (Standard Quality)
QQ-S-637	Steel, Bar, Carbon, Cold Finished (Standard Quality, Free Machining)
QQ-S-763	Steel Bars, Shapes and Forgings, Corrosion Resisting
QQ-Z-325	Zinc Plating (Electrodeposited)

MILITARY

MIL-C-5	Capacitors, Fixed, Mica-Dielectric
MIL-R-11	Resistors, Fixed, Composition (Insulated)
MIL-M-14	Plastic-Materials, Molded and Plastic Parts, Molded; Thermosetting
MIL-R-26	Resistors, Fixed, Wire-Wound, (Power Type)
MIL-T-27	Transformers and Inductors (Audio, Power and High Power Pulse)
MIL-P-79	Plastic-Materials, Laminated, Thermosetting; Rods and Tubes
MIL-P-116	Preservation, Methods of
MIL-B-857	Bolts, Nuts, Studs, and Tap Rivets (and Material for Same)
MIL-S-867	Steel, Castings, Corrosion-Resisting and Austenitic
MIL-R-900	Rubber, Synthetic, Medium-Soft, Molded, Sheet and Strip (for Watertight and Airtight Closures)
MIL-R-1149	Rubber, Synthetic, Gaskets, Packing, Sheets and Strip
MIL-M-3171	Magnesium, Alloy, Process for Corrosion, Protection of
MIL-C-5015	Connectors; Electrical, "AN" Type
MIL-P-6808	Primer Coating, Zinc Chromate, for Aircraft Use
MIL-A-8625	Anodic, Coatings, for Aluminum and Aluminum Alloys
MIL-R-15624	Rubber Material, Synthetic, Medium Soft; (Sheet, Strip, and Molded Shapes)
MIL-B-16540	Bronze, Phosphor; Castings
MIL-W-16878	Wire, Electrical Insulated, High Temperature
MIL-A-21180	Aluminum-Alloy Castings
MIL-I-24092	Insulating Varnish, Electrical, Impregnating

STANDARDS

MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristcs Requirements for Equipment

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MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MS 3102	Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type
MS 3106	Connector, Plug, Electric, Straight, Solder Contacts, AN Type

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CODE IDENT. 10001)

LD 495868	Shipping (Packing) Condition
Dwg 815483	Transformer
Dwg 877433	High Voltage Power Supply Assembled
Dwg 1803486	Shipping (Packing) Condition

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

WR-11	Design and Test of Packaging, Packing, Shipping and Handling Equipment for Weapon System Components
WR-43	Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, NW, Washington, D.C. 20036.)

3 REQUIREMENTS

3.1 Description. The high voltage power supply is a device which transforms 780 volts root mean square (rms), 3 phase wye, 400 hertz (Hz) into two dc voltages for use by the torpedo electronic panel. The 780 volts rms, 3 phase wye power is taken from a remotely located

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transformer which draws its power from the torpedo 115 volt rms, 400 Hz, 3-phase power lines. The high voltage power supply is required to furnish two off-on repeat-cycle loads.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the power supply shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. See 6.2.1.

3.3 Design objectives. The basic design objectives are that the equipment will meet the needs of the Naval service and that the final product will reflect the utmost in simplicity, have maximum reliability consistent with the state of the art, be easy to operate and maintain, and be economical to produce.

3.3.1 Miniaturization. It is intended that equipments use minature parts throughout, insofar as is practicable. The parts shall be in accordance with applicable specifications where possible. However, the use of parts not in actordance with an approved applicable specification will be given consideration provided that:

(a) It can be demonstrated to the procuring activity or appointed agent that there will be an appreciable reduction in weight, space, or both.

(b) Tests have been performed which demonstrate that the performance requirements for the equipment, particularly with respect to life, temperature, and overload, will be fulfilled.

(c) An adequate source of such parts is available.

3.4 Reliability. Reliability shall be in accordance with MIL-STD-454, Requirement 35.

3.5 Safety. The equipment shall have provisions for personnel safety in accordance with MIL-STD-454, Requirement 1.

3.6 Accessibility. Accessibility shall be in accordance with MIL-STD-454, Requirement 36.

3.7 Maintainability. Maintainability shall be in accordance with MIL-STD-454, Requirement 54.

3.8 Test points. Test points shall be provided in readily accessible locations for measuring, observing, or injecting those quantities pertinent to maintenance and repair of the equipment. Test provisions shall be in accordance with MIL-STD-454, Requirement 32.

3.9 Parts and materials, general. Materials and parts used in the construction of equipment shall be of a quality consistent with the proposed use and specified performance of the equipment. The employment of design which will tend to minimize the size and weight of the complete equipment shall be a major consideration and shall be used to the maximum practicable extent.

3.9.1 Standard parts and materials. The greatest practicable use commensurate with design requirements shall be made of standard parts and materials. Standard parts and materials are defined as those covered by: (1) Documents invoking or invoked by this specification; or (2) military, federal, or DOD-approved commercial specifications or standards. The use of other parts shall be in accordance with MIL-STD-454, Requirement 22.

3.9.2 Mercury and silicone. Materials capable of giving off either mercury or silicone vapor when subjected to temperatures within the operating and storage range of the high voltage power supply shall not be used.

3.9.3 Fungus and corrosion resistance. Parts and materials shall be fungus inert and corrosion resistant in accordance with MIL-STD-454, Requirements 4 and 15.

3.10 Parts, general.

3.10.1 Threaded parts. Bolts, nuts, screws, and studs shall conform to the requirements of FF-S-85, FF-S-86, FF-S-92, FF-N-836, or MIL-B-857. Screw threads for all threaded parts shall be in accordance with Screw Thread Standards for Federal Services, Handbook H28, and shall, except where necessitated by the exigencies of design, be of the coarse thread series. Threaded

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parts (other than bolts, studs, screws, and nuts) requiring threads of special diameter and pitch, such as for use on thin walled tube screw covers or plugs, shall employ the special pitch of 16 threads to the inch. For applications where a finer pitch than 16 threads to the inch is necessary, one of the following shall be selected: 20, 28, 36, 44, 56.

3.10.1.1 Materials. All screws, studs, spade bolts, and nuts shall be of brass or non-ferrous material of equal strength and suitaby protected against corrosion. When required for reasons of strength, corrosion-resistant steel shall be used. Threaded inserts for aluminum parts shall be made of corrosion-resistant steel, threaded inserts for molded or laminated plastics shall be made of brass or corrosion-resistant steel. Aluminum nuts shall not be used.

3.10.1.2 Thread projection. Screws secured by nuts or other retaining devices shall be of such length that there shall project beyond the retainer a minimum of 1-1/2 threads and a maximum of 1/8 inch (3mm) plus 1-1/2 threads for screws up to one inch in length and 1/4 inch (6mm) plus 1-1/2 threads for screws over one inch in length. These requirements shall not be construed as precluding the use of screws assembled in blind holes, spacers, and similar usage, or where the design restrictions require that the threaded portion of the screw be flush with the retaining device. In no case shall screws be clipped, cut, or otherwise shortened after plating. See 6.4.

3.10.1.3 Self-tapping screws. Self-tapping screws shall not be used except where employed in approved components and where their removal from the component is a remote possibility.

3.10.1.4 Locking devices. Locking devices shall be provided for all screw-thread assemblies except for captive screws on withdrawable units. The working loose during tests or in service of threaded parts will be considered a defect in design. All lockwashers shall be fabricated of corrosion-resistant steel, plated phosphor bronze, or zinc plated steel in accordance with QQ-Z-325, Type II, Class 2.

(a) External-tooth type lockwashers shall not be used with screws 1/4 inch (6mm) or greater in diameter, or for screws on external surfaces.

(b) Spring type lockwashers may be used for screws of all sizes.

(c) Internal-tooth lockwashers shall be used for screws and for other special application where space limitation or appearance necessitate their use. They shall not be used under binder head screws.

(d) Screws with captive lockwashers may be used provided that both the screws and lockwashers conform to Government specification requirements with respect to types, materials, and applications, and that replacement with corresponding or equal parts can be effected in the field.

(e) Flat-head screws threaded into tapped holes shall be coated with alkyd resin varnish.

3.10.2 Springs. Springs shall be in accordance with MIL-STD-454, Requirement 41.

3.11 Materials, general.

3.11.1 Aluminum alloys. Aluminum alloys used shall conform to QQ-A-250/3, QQ-A-250/8, QQ-A-250/11, QQ-A-367, or MIL-A-21180.

3.11.2 Steel.

3.11.2.1 Steel, corrosion-resistant. Steel alloys of the corrosion-resistant type shall be used in applications where strength requirements indicate the use of steel. These alloys shall conform to QQ-S-763 or MIL-S-867.

3.11.2.2 Steel, other types. Where the use of corrosion-resistant steel is not practical, steel alloys used shall conform to QQ-S-630, QQ-S-634, or QQ-S-637.

3.11.3 Brass. Brass shall conform to QQ-B-613, QQ-B-626, QQ-B-637, or QQ-C-390.

3.11.4 Bronze. Bronze shall conform to QQ-C-390, QQ-C-450, QQ-B-750, or MIL-B-16540.

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3.11.5 Electrical insulating materials. Electrical insulating materials shall be in accordance with MIL-STD-454, Requirement 11.

3.11.6 Plastics.

3.11.6.1 Plastics, laminated, thermosetting. Laminated thermosetting plastic materials shall conform to L-P-387 or MIL-P-79. Parts fabricated from materials intended for use as electrical insulation shall be vacuum impregnated with a suitable varnish, such as one meeting the requirements of MIL-I-24092, and dried, after all machining and punching operations have been completed.

3.11.6.2 Plastics, molded, thermosetting. Molded thermosetting plastic materials shall conform to the requirements of MIL-M-14. Machining operations of parts molded from such material shall be held to a minimum. Parts which have been machined shall be subsequently vacuum impregnated with a suitable varnish, such as one meeting the requirements of MIL-I-24092, and dried.

3.11.7 Rubber. Rubber for gaskets shall conform to MIL-R-900, MIL-R-1149, or MIL-R-15624.

3.12 Electronic parts.

3.12.1 Selection. All electronic parts shall be selected as to class, type, size, rating, operating temperatures, and intended use, and so located as to not become damaged or have their useful life seriously shortened when the equipment is subjected to the conditions specified herein or to any combinations thereof. No part shall contain sealing or insulating compounds which will crack, soften, or flow under any of the conditions specified herein, unless confined in such a manner that the compound will not flow from the container.

3.12.2 Interchangeability. Interchangeability shall be in accordance with MIL-STD-454, Requirement 7. Parts selected shall require a minimum number of different types or ratings, and shall be interchangeable physically and electrically with corresponding (but not necessarily identical) parts employed in the equipment and furnished as spare parts, without resorting to drilling, filing, and similar operations, or the use of undue force. Due precaution shall be exercised to provide for design tolerances, especially as to mounting dimensions sufficient to accommodate any replacement part bearing the same type designation as the replaced part and having the maximum physical dimensions permitted for that type designation. This shall be accomplished without adversely affecting the specified part or the equipment performance. Except for the shortening of lead lengths, alterations to or modifications of standard parts shall not be made.

3.12.3 Derating. Derating shall be in accordance with MIL-STD-454, Requirement 18. Parts shall be located so that the working temperature, under full load operating conditions, does not exceed the maximum temperature prescribed by specifications for the part.

3.12.4 Capacitors. Capacitors shall be in accordance with MIL-STD-454, Requirement 2.

3.12.4.1 Capacitors, fixed, mica-dielectric. Fixed, mica-dielectric capacitors for use at frequencies up to three magacycles shall conform to the requirements of MIL-C-5, except that capacitors conforming to characteristics "A", as defined therein, shall not be used. For potted type fixed mica-dielectric capacitors, the actual measured radio frequency current load imposed on the capacitors and the corresponding peak working voltage across the capacitors shall be measured or calculated and shall not exceed the rated values listed in MIL-C-5.

3.12.4.2 Capacitors connected in parallel or series shall not be used in lieu of a single capacitor having the correct rating.

3.12.4.3 Capacitors shall be selected from the nominal values and tolerances listed in the applicable part specifications. Capacitors having the widest tolerance of capacitance and temperature coefficient shall be used, and circuits shall be so designed, insofar as is practicable.

3.12.5 Resistors. Resistors shall be in accordance with MIL-STD-454, Requirement 33.

3.12.5.1 Resistors, fixed composition. Composition type fixed resistors shall be of the insulated type and conform to the requirements of characteristics "BF" and the other applicable requirements of MIL-R-11.

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3.12.5.2 Nonapproved types. The following types of fixed composition resistors shall not be used:

RC15, RC16, RC25, RC35, RC38, RC45, RC65,
RC75, RC76, (See MIL-R-11).

3.12.5.3 Application. Particular care shall be exercised in the application of fixed composition resistors to ensure that for each particular resistor application and under any specified condition of line voltage variation, exclusive of surges, rated maximum continuous working voltage (RCWV) will not be exceeded.

3.12.5.4 Resistors, fixed, power, wire-wound. Fixed, wire-wound power resistors shall conform to the requirements of characteristics "F", Grade 1, Class 1, and the other applicable requirements of MIL-R-26. When the use of resistors having characteristics other than "F" is permitted by the individual equipment specification, the selection of such resistors shall be made in the following order of preference:

Characteristics	G - V

3.12.5.5 Resistors connected in parallel or series shall not be used in lieu of a single resistor having the correct rating.

3.12.5.6 Resistors shall be selected from the nominal values and tolerances listed in applicable part specifications. Resistors having the widest tolerance of resistance and temperature coefficient shall be used, and circuits shall be so designed, insofar as is practicable.

3.12.6 Semiconductor devices. Semiconductor devices shall be in accordance with MIL-STD-454, Requirements 30 and 64. The maximum reverse breakdown voltage of each individual rectifier used shall be at least 3000 volts.

3.12.7 Hookup wire. The hookup wire for internal wiring shall be in accordance with MIL-W-16878, except that wire having extruded insulation shall be used in lieu of braided insulated wire wherever possible. All hookup wire shall be stranded except where otherwise required in radio frequency circuits.

3.12.7.1 Color code. All wire used in the wiring of the equipment shall be distinctly color coded. The system of color coding shall be such that circuit tracing can be effected by servicing personnel.

3.12.8 Connectors, Type "AN". Type "AN" connectors for multi-conductor cables shall be in accordance with MIL-C-5015. The use of die cast bodies for these connectors is permitted, providing optional types fulfilling the material requirements specified elsewhere herein are not available.

3.12.9 Inductors. Inductors shall conform to the requirements of Grade 2, Classes A, B, or C of MIL-T-27.

3.12.9.1 Coil preparation and impregnation. Coils for all transformers, inductors, and chokes shall be dried and thoroughly impregnated, except for coils that are contained in hermetically sealed cases. The insulation of the coil is considered impregnated when a suitable substance does not completely fill the spaces between the insulated conductors. The impregnating substance, in order to be considered suitable, must have good insulating properties, must entirely cover the fibers and render them adherent to each other and to the conductor, must not produce interstices within itself as a consequence of the solvent or through any other cause, and must not flow, crack, or unduly deteriorate during the operation of the transformer or inductor at the operating temperatures specified herein.

3.12.9.2 Potting. Filling, potting, or sealing compounds, liquid or solid, shall close all voids and air spaces. Previous to sealing, potting, or filling, the coils shall be treated to remove all moisture and foreign substances.

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3.12.9.3 Connection to terminals. Connections of terminals to windings shall result in joints mechanically and electrically permanent and shall prevent the entrance of moisture into the windings directly or by wick action of the lead insulation.

3.13 Processes.

3.13.1 Corrosion. Contacts between dissimilar metals shall be suitably treated to inhibit electrolytic action. Requirements contained in various paragraphs of this specification are intended to insure a high degree of corrosion resistance. However, it shall be incumbent on the contractor to provide additional protection from corrosion in such cases where experience or judgment indicates that such added protection is required.

3.13.2 Coatings. All parts manufactured from materials not resistant to corrosion shall be provided with a protective coating or finish, as specified herein.

3.13.2.1 Anodic films. Parts fabricated from aluminum alloys shall be "anodized" in accordance with MIL-A-8625.

3.13.2.2 Ferrous metal parts. Ferrous metals subject to corrosion, if used, shall be zinc plated in accordance with QQ-Z-325, Type II, Class 2, if such treatment has no deleterious effect on electrical performance.

3.13.2.3 Magnesium alloy parts. Magnesium and magnesium alloys, if used, shall be treated in accordance with Specification MIL-M-3171, Type III, followed by priming in accordance with MIL-P-6808.

3.13.2.4 Plating. Plating of metal surfaces shall be in accordance with the requirements of the applicable specification for each type as follows:

(a) Chromium:	QQ-C-320
(b) Silver:	QQ-S-365
(c) Cadmium:	QQ-P-416-Type II, Class A
(d) Zinc:	QQ-Z-325-Type II, Class 2
(e) Nichel:	QQ-N-290

Cadmium plating shall not be used in confined enclosures where no air is circulated and when materials are present from which chlorine may be liberated, such as chlorinated rubber and polyvinyl chlorine, or when the flaking or corrosion by-products of the cadmium may cause mal-operation of the equipment.

3.13.3 Soldering. Solder and soldering shall be in accordance with MIL-STD-454, Requirement 5.

3.13.4 Electronics parts mounting. Electronic parts shall not be mounted on other electronic parts. In no case shall parts weighing more than 1/2 ounce (14g) be supported solely by their leads or terminals. All parts weighing 1/2 ounce (14g), or less, shall be mounted on mounting boards, one part soldered to each pair of terminal posts with the body of the part in intimate contact with the surface of the mounting board. Each part lead shall be not less than 1/4 inch (6mm) and not more than 1/2 inch (13mm) in length and shall not be bent closer than 1/8 inch (3mm) to the body of the part. Mounting and terminal boards may be installed vertically or horizontally depending on the available space, and shall be so mounted that the back surfaces thereof are spaced a distance of not less than 1/16 inch (1.5mm) from the mounting surface. It shall be possible to remove or replace any part and wiring from the mounting boards and wiring from terminal boards without removing the boards from their supports or causing a major disturbance of wiring and parts not directly connected to the part.

3.13.5 Methods of wiring.

3.13.5.1 Cabling of leads. Individual wires shall be arranged, where practical, so as to permit grouping into prefabricated cables. Wires, when cabled together, shall be laced with acceptable cord. Prefabricated cables and branches thereof shall be securely clamped to the chassis (twine or tape shall not be used for this purpose).

3.13.5.2 No more than three wires, including wires from parts, shall terminate at any one terminal.

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3.13.5.3 Pressure contacts. Electrical connections shall not depend upon wires, lugs, terminals, and the like clamped between a metallic and an insulating or pliable material. Connections of this type shall be clamped between rigid metal members, such as in an assembly of a machine screw, two nuts, and two washers. If such an assembly cannot be used, and maintenance of a tight connection depends upon the resistance of an insulating or pliable material to compressive stress or shear, the connection shall be securely soldered.

3.13.6 Marking.

3.13.6.1 Parts. The marking of parts shall be in accordance with the applicable individual specifications listed herein.

3.13.6.2 Assembly. The high voltage power supply shall be marked as specified below, and on Drawing 877433.

(a) Caution note. The characters in the word "CAUTION" shall be at least 1/4 inch (6mm) high; remaining characters in the note shall be at least 1/8 inch (3mm) high.

(b) Wiring diagram. A wiring diagram complete with legend describing the electrical components of the equipment shall be attached to the high voltage power supply. When possible the wiring diagram shall be placed on the protected side of a removable cover plate.

3.13.7 Welding. All welds shall be of ample size, with thorough penetration and good fusion, and shall be substantially free of injurious scabs, blisters, abnormal pockmarks, bruises, or other surface defects. Welds shall be capable of withstanding a strength test to destruction in which failure shall not occur within itself. There shall be no evidence of harmful effects due to the welding process on material adjacent to the weld. All welds shall be cleaned of scale, oxidation products, and excess flux by sand blasting, wire brushing, buffing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed. Where tack or spot welds are employed the number of welds shall be sufficient to provide adequate strength; at least two welds shall be used to hold any part.

3.14 Physical requirements.

3.14.1 Weight. The weight of the high voltage supply shall not exceed 20 pounds (9.07kg).

3.14.2 Dimensions. The mounting and outside dimensions of the assembled high voltage supply shall be as specified on Drawing 877433.

3.15 Performance requirements and product characteristics. The high voltage power supply shall meet the following performance requirements and product characteristics:

3.15.1 Input power. The high voltage power supply shall meet all performance requirements and product characteristics when supplied with input power as follows:

(a) Voltage: 780 ± 39 volts (rms) AC, line to neutral, 3-phase, wye-connected, measured at the input terminals of the high voltage power supply.

(b) Frequency: 400 ± 20 Hz.

(c) Source impedance: 196 ohms ± 15 percent (%) per phase at 400 Hz.

3.15.2 Input current. Exclusive of starting surge, the high voltage power supply input current shall not exceed 0.350 ampere peak-to-peak per line.

3.15.3 Output connections. Two dc output circuits shall be provided as indicated on the receptacle wiring diagram of Drawing 877433. The circuit connected to terminals G and E of receptacle MS-3102A-24-27P shall be referred to hereinafter as circuit 1. The circuit connected to terminals F and E shall be referred to hereinafter as circuit 2.

3.15.4 Open circuit output characteristics.

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3.15.4.1 DC voltage.

Circuit 1:	2000 volts dc maximum
Circuit 2:	850 volts dc maximum

3.15.4.2 AC voltage content. The total ac voltage content in the open circuit output voltage of either circuit shall not exceed 20 millivolts rms in the frequency band from 55 to 65 kHz.

3.15.4.3 Radiation. When operated under open circuit conditions, the high voltage power supply shall meet the radiation limits specified in MIL-STD-461, Table A IV, Category SB2.

3.15.5 Loaded output characteristics.

3.15.5.1 Load cycles. The high voltage power supply shall meet the performance requirements specified herein when the output is loaded by load cycle 1 or load cycle 2, defined as follows:

(a) Load cycle 1. Load shall be applied simultaneously to both output circuits once during each 1250 ± 8 milliseconds (ms) cycle of operation for a period of 46 ± 3 ms. The start of the 46 ± 3 ms load period shall define the start of each 1250 ± 8 ms cycle.

(b) Load cycle 2. Load shall be applied simultaneously to both output circuits once during each 625 ± 4 ms cycle of operation for a period of 46 ± 3 ms. The start of the 46 ± 3 ms load period shall be defined as the start of each 625 ± 4 ms cycle.

3.15.5.2 Circuit loading. The high voltage power supply shall meet the performance requirements specified herein with circuit 1 loaded by 1200 ohms ± 1%, resistive, and with circuit 2 loaded by 3060 ohms ± 1%, resistive; as described in load cycles 1 and 2.

3.15.5.3 Output characteristics during load cycle 1.

Circuit 1: 1800 ± 180 VDC peak voltage and 650 volts maximum decay from the peak voltage during the load period.

Circuit 2: 650 ± 65 VDC peak voltage and 175 volts dc maximum decay from the peak voltage during the load period.

3.15.5.4 Output characteristics during load cycle 2.

Circuit 1: 1550 ± 155 VDC peak voltage and 575 volts dc maximum decay from the peak voltage during the load period.

Circuit 2: 500 ± 50 VDC peak voltage and 200 volts dc maximum decay from peak voltage during the load period.

3.15.6 Output impedance. The output impedance of the high voltage power supply at a frequency of 60 kHz shall not exceed the following limits:

Circuit 1:	3.5 ohms maximum
Circuit 2:	7.0 ohms maximum

3.15.7 Safety requirements. Within 10 ± 1 minutes after maximum input voltage has been removed, the open circuit output voltages of the high voltage power supply shall not exceed the following:

Circuit 1:	75 volts dc
Circuit 2:	10 volts dc

3.15.8 Dielectric requirements. The high voltage power supply shall be capable of withstanding for one minute 2900 volts rms at 60 Hz applied between all pins of the receptable MS-3102A-24-27P simultaneously and the chassis without any arcing or breakdown of insulation.

3.15.9 Insulation resistance. The insulation resistance at 500 volts dc shall be 100 megohms minimum between all pins of receptacle MS-3102A-24-27P simultaneously and the chassis.

3.15.10 Operating time

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3.15.10.1 Open circuit. The high voltage power supply shall be capable of being left energized for periods of 9 hours and 50 ± 10 minutes before the repeat-cycle load is applied.

3.15.10.2 Repeat-cycle load. The high voltage power supply shall be capable of operating for 30 ± 5 minutes under either load cycle 1 or load cycle 2, followed by sufficient time to cool to ambient temperature.

3.15.11 Operating life. The high voltage power supply shall be capable of operating either under open circuit or repeat-cycle load conditions without failure for a total of 200 hours minimum.

3.15.12 Surge capability. The high voltage power supply shall be capable of being switched on and off at a rate of 20 pulses per minute for a period of one minute as specified in 4.10.3 without failure.

3.15.13 High voltage capability. The high voltage power supply shall be capable of withstanding a voltage of 4000 volts dc between pin G and pin E of receptacle MS-3102A-24-27P and case and a voltage of 1700 volts dc between pin F and pin E of receptacle MS-3102A-24-27P and case without arcing or insulation breakdown.

3.15.14 Receptacle connections. The pins of high voltage power supply receptacle MS-3102A-24-27P shall be utilized as specified on Drawing 877433.

3.15.15 Grounds. No circuit within the high voltage power supply shall be connected to the high voltage power supply chassis.

3.16 Environmental requirements.

3.16.1 Vibration, nonoperating. The high voltage power supply, less pack and packaging, shall be capable of withstanding the vibrations of Schedule I, specified in 4.10.12.1, within the frequency range of 10-100 Hz in each of three mutually perpendicular axes without damage, and shall remain capable of meeting all the performance requirements and product characteristics of this specification.

3.16.2 Vibration, operating. The high voltage power supply shall be capable of withstanding the vibrations of Schedule II, specified in 4.10.12.2, within the frequency range of 20-150 Hz in each of three mutually perpendicular axes. During and after such vibration, the high voltage power supply shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.16.3 Shock; operating condition. The high voltage power supply, mounted as specified in 4.4.2, shall withstand 3 shock pulses in each of the 3 mutually perpendicular planes. The shock pulse shall have a peak magnitude of 60 ± 6 g's, be not less than 8 ms in duration, and shall approximate a one-half sine wave. Thereafter, the high voltage power supply shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.16.4 Acceleration. The high voltage power supply shall withstand 15 ± 2 g's acceleration for a period of not less than 0.2 second in the direction of the longitudinal axis, with inertia load aft, without damage and shall meet all the performance requirements and product characteristics of this specification.

3.16.5 Temperature and humidity.

3.16.5.1 High temperature and humidity, nonoperating. The high voltage power supply, less pack and packaging, shall be capable of withstanding a temperature of 71° ± 3°C (+ 160° ± 5°F) for a period of 48 hours at a relative humidity of 90 to 95 percent without damage and shall remain capable of meeting all the performance requirements and product characteristics of the specification.

3.16.5.2 High temperature, operating. The high voltage power supply shall be capable of meeting the performance requirements of this specification while being maintained at a temperature of 54°C (130°F).

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3.16.5.3 Low temperature, nonoperating. The high voltage power supply, less pack and packaging, shall be capable of withstanding a temperature of -54° ± 3°C (-65° ± 5°F) for a period of 48 hours. Thereafter, the unit shall meet all the performance requirements and product characteristics of this specification.

3.16.5.4 Low temperature, operating. The high voltage power supply shall be capable of meeting the performance requirements of this specification while being maintained at a temperature of -2°C (28°F).

3.16.6 Pressure. The high voltage power supply, while in the nonoperative condition, shall be capable of withstanding 180 external pressure cycles ranging from room ambient to 4 psia + 0 -2 psia (28 + 0, -14kPa) (vacuum) in accordance with test procedures specified in 4.4.6, followed by one 30-minute period 4 psia + 0, - 2 psia (28 + 0, -14kPa) (vacuum) test. See 6.4. Thereafter, the power supply shall show no evidence of damage or oil leakage and shall meet the performance requirements and product characteristics of this specification.

3.16.7 Transportation environmental requirements.

3.16.7.1 Vibration; transportation (Level C only). The high voltage power supply, packaged and packed in accordance with Section 5 or the contract or purchase order, shall withstand the vibration specified in 4.4.11. Thereafter, the power supply shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 6.2.1.

3.16.7.2 Shock; transportation (Level C only). The high voltage power supply, packaged and packed in accordance with Section 5 or the contract or purchase order, shall withstand the shocks specified in 4.4.12. Thereafter, the power supply shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 6.2.1.

3.17 Workmanship.

3.17.1 General. The equipment, including all parts, shall be manufactured and finished in a thoroughly workmanlike manner. Particular attention shall be paid to reliability, safety, and appearance. Where dimensions and tolerances affect interchangeability, operation, or performance, the tolerance shall be applied and held, so that selective fitting shall not be required. All wiring shall be arranged in a neat and workmanlike manner.

3.17.2 The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

3.17.3 Screw assemblies. Assembly screws shall be tight. The word "tight" means that the screw cannot be appreciably tightened further without damage or injury to the screw threads or assembly.

3.17.4 Riveting and swaging. Riveting and swaging operations shall be carefully performed to insure that the rivets and terminals are tight and satisfactorily headed.

3.18 Cleanup. Units shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material after assembly. Burrs and sharp edges, as well as rosin flash which might crumble, shall be removed.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major, or Minor, in accordance with WR-43. Tests and

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examinations that verify critical characteristics are identified by the symbol (C), and those that verify major characteristics are identified by the symbol (M). The number following the classification symbol is the serial number of the test or examination. Tests and examinations that are not annotated with a classification code are classified minor. See 6.2.1.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. Preproduction Inspection (see 4.6)
2. Quality Conformance Inspection (see 4.7)
3. Periodic Production Inspection (see 4.8)

4.4 Inspection conditions. Unless otherwise specified, the high voltage power supply shall be subjected to inspections under the following conditions:

(a) Temperature: 18° to 35°C (65° to 95°F)
(b) Altitude: normal ground level
(c) Humidity: room ambient to 95 percent relative
(d) Pressure: normal atmospheric

4.5 Inspection equipment. The following items are required to perform the quality conformance inspection specified herein.

4.5.1 AC power supply with the following characteristics:

(a) Voltage: 780 volts rms line-to-neutral, 3 phase wye, with provisions for varying the line-to-neutral voltage from 740 to 820 volts rms.

(b) Frequency: 400 ± 20 Hz

(c) Capacity: Not less than 0.200 ampere peak per line under continuous operation. Inrush capacity shall be not less than 0.270 ampere peak per line.

(d) Output impedance: 196 ± 29.4 ohms per phase at 400 Hz.

4.5.2 AC power supply, 115 V ± 3%, 400 ± 8 Hz, 3-phase.

4.5.3 Auxiliary supplies: Auxiliary supplies will be required to operate the load relay and the repeat-cycle timer listed below. The voltage and capacity of these supplies will be dictated by the type of load relay and repeat-cycle timer used.

4.5.4 Transformer in accordance with Drawing 815483.

4.5.6 AC voltmeter capable of reading 820 volts rms at 400 Hz with 0.3 percent accuracy; input impedance: one megohm minimum shunted by 100 picofarads maximum.

4.5.7 AC voltmeter capable of reading 0.02 and 0.2 volt rms at 60 kHz with 2% accuracy, minimum input impedance shall be 60 kilohms or 100 times the terminating or output impedance of the bandpass filter listed in 4.5.15, whichever is greater.

4.5.8 DC voltmeter capable of reading 2000 and 850 volts dc with 1% accuracy; two megohm minimum input resistance.

4.5.9 Calibrated oscilloscope. The oscilloscope will be used to measure the magnitude and duration of the repeat-cycle output voltages furnished by the high voltage power supply to the dummy loads specified below. Since the circuit 1 output voltage is higher than the allowable input voltage for most commerically available oscilloscopes, the repeat-cycle output voltages may be measured by connecting the oscilloscope vertical deflection input terminals between any convenient point on either dummy load and the common zero volt dc terminal of the two dc output circuits. If voltage measurements are made as suggested above, the tap point resistance shall be compared to the total resistance of the applicable dummy load so that suitable correction factors may be applied to the measured voltage to determine actual output voltages. The following voltage requirements apply to an oscilloscope, the input terminals of which would be connected across the entire dummy load in each instance:

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(a) Range of measurement: 2, 850, and 2000 volts full scale with a minimum full scale deflection of 3 inches (75 mm).

(b) Frequency response: flat, dc to 100 kHz. Down not more than 10 percent at 100 kHz.

(c) Sweep: Externally driven with allowable triggering voltage of the magnitude required to operate the load relay specified below. Sweep range shall be such that a 50 millisecond occurrence can be fully displayed.

(d) Input impedance: One megohm shunted by 100 picofarads maximum.

4.5.10 Low frequency oscillator: The low frequency oscillator shall be capable of furnishing 10 ± 5 volts rms at 60 Hz when supplying a 685 ohm resistive load.

4.5.11 Load relay: A dc operated relay with a minimum of two sets of normally open contacts of a capacity sufficient to accomplish the following:

(a) Close a 1.7-ampere resistive load at 2000 volts dc, and open a 1.3-ampere resistive load at 1500 volts dc once per 1.25 seconds for 30 ± 5 minutes.

(b) Close a 1.5-ampere resistive load at 1700 volts dc, and open a 1.1-ampere resistive load at 1300 volts dc once per 0.625 second for 30 ± 5 minutes.

(c) Limit relay contact bounce on closing to not more than two milliseconds.

4.5.12 Repeat-cycle timer: The repeat-cycle timer will be used to control the action of the specified load relay so that it operates once for a given period during each cycle. The timer shall consist of (1) at least one switch of sufficient capacity to operate the load relay; (2) an adjustable cam or cams necessary to operate the switch; and (3) a motor to drive the cam shaft. The repeat-cycle timer shall have the following capabilities:

(a) It shall be possible to set the timer for a cycle time of either 1250 ± 8 ms or 625 ± 4 ms.

(b) When the timer is set for "load cycle 1" the timer switch shall make one complete operation each 1250 ± 8 ms. The start of the cycle shall occur when the switch closes. The switch shall remain closed long enough so that the load relay completes the high voltage supply output circuits for 46 ± 3 ms.

(c) When the timer is set for "load cycle 2", the timer switch shall make one complete operation each 625 ± 4 ms. The start of the cycle shall occur when the switch closes. The switch shall remain closed long enough so that the load relay completes the high voltage supply output circuits for 46 ± 3 ms.

4.5.12.1 A three phase interrupter shall be included that shall be capable of being on for 1.50 seconds and off for 1.50 seconds. The combined transfer time and tolerance for each "on" and "off" period shall be no greater than 4 milliseconds. The interrupter shall be capable of withstanding 300 milliamperes, peak.

4.5.13 Dummy loads.

4.5.13.1 Circuit 1: A 1200-ohm, ± 1 percent, 150 watt, non-inductive, resistive load capable of withstanding 2000 volts dc.

4.5.13.2 Circuit 2: A 3060-ohm, ± 1 percent, 10 watt, non-inductive, resistive load capable of withstanding 850 volts dc.

4.5.14 Miscellaneous resistors.

4.5.14.1 Shunt: 10 ohms, ± 1 percent, 1/2 watt, non-inductive. This resistor shall be used as an input current shunt.

4.5.14.2 Oscillator: 680 ohms, ± 10 percent, 1/2 watt, non-inductive; and 5 ohms, ± 1 percent, 1/2 watt, non-inductive. The 5 ohm resistor will be used as an impedance reference.

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4.5.15 Bandpass filters: The bandpass filter shall be used during the measurement of ac voltage content in the high voltage power supply open circuit output voltages in a frequency band from 55 to 65 kHz. The purpose of this filter shall be to transfer maximum voltage, not maximum power, in its pass band. Since the ac output impedance of each high voltage power supply output circuit is relatively low, the inclusion of a simulated generator impedance ahead of the filter shall be allowed if corrections are applied to the test measurements to account for the effect of this impedance. The passband filter shall have the following characteristics:

(a) Cut-off frequencies: 55 ± 1 kHz and 65 ± 1 kHz.

(b) Voltage attenuation in pass band: 0 ± 3 db from 55 to 65 kHz.

(c) Voltage attenuation in rejection regions: 18 db per octave reference, with maximum attenuation of at least 80 db.

(d) Input impedance: The minimum input impedance of the filter including the simulated generator impedance, if used, shall be not less than 7000 ohms from 55 to 65 kHz.

(e) DC voltage: The filter shall be capable of withstanding 2000 volts dc maximum continuously applied to its input terminals.

(f) AC voltage: The filter components shall be capable of withstanding up to one volt ac rms on the filter input terminals.

4.5.16 Connector: MS-3106A-24-27S.

4.5.17 High potential test set: The high potential test set shall be capable of furnishing 2900 volts rms at 60 Hz. The test set will be used for insulation testing and shall include an alarm bell or light, or combination of both, to indicate when arcing or breakdown occurs.

4.5.18 Three pole switch: The three pole, single throw switch shall be capable of withstanding 2900 volts rms at 60 Hz for one minute applied between each pair of adjacent poles without arcing or breakdown of insulation. Each switch pole shall be capable of carrying and breaking 0.200 ampere resistive load current at 780 volts rms without any evidence of overheating. Maximum contact drop at 0.200 ampere shall not exceed 0.5 volt rms.

4.6 Preproduction inspection.

4.6.1 Preproduction samples. Unless otherwise specified in the contract or order, a sample of five high voltage power supplies shall be manufactured using the methods, materials, and procedures proposed for production lots. Further production by the contractor prior to approval of the samples shall be at the contractor's risk. See 6.2.1.

4.6.2 Preproduction inspection procedure. Each preproduction sample shall be subjected to the preproduction inspection specified in Table I. See 6.2.2.

4.6.3 Preproduction failures. The failure of any preproduction sample to pass any preproduction test or examination shall be cause for, and may result in, denial of production approval.

4.7 Quality conformance inspection.

4.7.1 Quality conformance samples. Each high voltage power supply shall be a quality conformance sample. (100% inspection)

4.7.2 Quality conformance inspection procedure. Each high voltage power supply submitted for acceptance shall be subjected to the quality conformance inspection specified in Table I. See 6.2.2.

4.7.3 Quality conformance failures. The failure of any high voltage power supply to pass any quality conformance test or examination shall be cause for, and may result in, rejection of the defective item. See 6.2.1.

4.8 Periodic production inspection.

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4.8.1 Periodic production samples. Unless otherwise specified, a random sample of five high voltage power supplies shall be drawn from each succeeding lot of 300 items or less which has passed quality conformance inspection.

4.8.1.1 Lot size. As applied to inspection of units of production, the term "lot" shall mean "inspection lot," i.e., a collection of units of production submitted by a supplied for inspection. Unless otherwise specified, the number of units in inspection lots shall be 300 units. See 6.2.1.

4.8.2 Periodic production inspection procedure. Each periodic production sample shall be subjected to the periodic production inspection specified in Table I. See 6.2.2.

4.8.3 Periodic production failures. The failure of any periodic production sample to pass any periodic production test or examination shall constitute a defect. The consequences of a defect shall depend upon the classification of the characteristic being tested or examined (see 4.2), the specified number of acceptable defects, and other provisions of the contract or order. See 6.2.1. In general, a failure is cause for rejection of the lot the defective item represents.

4.9 Test setup. See 3.15.3 for circuit designations. Unless otherwise specified or not applicable, the following test setup shall be used for all operational tests and examinations specified herein.

4.9.1 Connect terminals B and C of connector MS-3106A-24-27S through the three pole switch (4.5.16) to two of the lines of the ac power source (4.5.1). Connect terminal A of the connector through the 10 ohm resistor (4.5.14.1) and the remaining pole of the three pole switch to the remaining line of the ac power source. Connect terminal D of the connector to the neutral point of the ac power source.

4.9.2 Connect terminal G (circuit 1) and terminal F (circuit 2) of connector MS-3106A-24-27S, each to one side of separate normally open contacts of the load relay specified in 4.5.11.

4.9.3 Connect the remaining circuit 1 relay contact to one side of the dummy load specified in 4.5.13.1 and connect the remaining circuit 2 relay contact to one side of the dummy load specified in 4.5.13.2.

4.9.4 Join the remaining terminals of both dummy loads together and connect them to terminal E of connector MS-3106A-24-27S. Plug the connector into the receptacle on the high voltage power supply with the three pole switch in the open position.

4.9.5 Connect the coil of the load relay in series with the repeat-cycle timer switch (4.5.12). This junction shall also be connected to the "external sync" terminal of the oscilloscope specified in 4.5.9.

4.9.6 Connect the remaining terminal of the load relay coil to the "ground" terminal of a suitable dc supply and the "ground" terminal of the oscilloscope. Connect the remaining repeat-cycle timer switch terminal to the "off ground" dc supply terminal. The repeat-cycle timer switch shall be in the open position so that the load relay is not energized.

4.10 Test and examination methods and procedures.

4.10.1 Preparation for delivery. The packaging, packing, and preparation for delivery shall be examined to determine conformance to the requirements of section 5 herein.

4.10.2 Materials, construction, and configuration. The high voltage power supply shall be visually examined and measured to determine conformance to the materials, construction and configuration requirements of 3.3 through 3.14, 3.15.3, 3.17, and 3.18 herein and applicable drawings.

4.10.3 Surge capability. (M101). Momentarily remove the connections from terminals A, B, and C of connector MS-3106A-24-27S, and connect these terminals to the secondary windings of the transformer of 4.5.4. Insert the interrupter of 4.5.12.1 between each transformer primary winding and each phase of a 115 volt, three phase power source. Apply this voltage for a period of not less than one minute. Upon completion of this test, remove the transformer and re-establish the connections specified in 4.9.1.

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WARNING

Use extreme care while testing the high voltage power supply. A potentially lethal level of energy could be stored in any output capacitors during and immediately after operation. At the conclusion of any test and before disconnecting the mating connector from the high voltage power supply receptacle, energize the load relay for five seconds to completely discharge any such capacitors.

Table I. Inspection Procedures

Test or Examination	Requirement	Method	Preproduction	Quality Conformance	Periodic Production
Preparation for Delivery	5	4.10.1	1[1]	1[1]	
Materials, Construction, Configuration	3.3 to 3.15 3.17, 3.18	4.10.2	2		1[1]
Setup		4.9	3	2	
Surge Capability	3.15.12	4.10.3	4	3	
Open Circuit Setup	3.15.4	4.10.4	5	4	
DC Voltage	3.15.4.1	4.10.4.1	6	5	
AC Voltage	3.15.4.2	4.10.4.2	7	6	
Safety	3.15.7	4.10.5	8	7	
Load Cycle 1	3.15.5.3	4.10.6	9	8	
Load Cycle 2	3.15.5.4	4.10.7	10	9	
Input Current	3.15.2	4.10.8	11	10	
Output Impedance	3.15.6	4.10.9	12	11	
Dielectric Strength	3.15.8	4.10.10	13	12	
Insulation Resistance	3.15.9	4.10.11	14	13	
Vibration Setup		4.10.12	15		2
Vibration, nonoperating	3.16.1	4.10.12.1	16		3
Vibration, operating	3.16.2	4.10.12.2	17		4
Vibration, transportation	3.16.7.1[2]	4.10.22[2]	18[2]		5[2]
Shock, operating	3.16.3	4.10.13	19		6
Shock, transportation	3.16.7.2[2]	4.10.23[2]	20[2]		7[2]
Acceleration	3.16.4	4.10.14	21		8
Temperature/Humidity	3.16.5.1	4.10.15	22		9
High Temperature	3.16.5.2	4.10.16.1	23		10
Low Temp., nonoperating	3.16.5.3	4.10.16.2	24		11
Low Temperature	3.16.5.4	4.10.16.3	25		12
Pressure (Vacuum)	3.16.6	4.10.17	26		13
Radiation	3.15.4.3	4.10.18	27		14
Operating Time	3.15.10	4.10.19	28		15
High Voltage	3.15.13	4.10.20	29		16
Operating Life	3.15.11	4.10.21	30		17

[1] Tests and Examinations shall be performed in increasing numerical sequence for each inspection classification, unless otherwise specified. See 6.2.1.

[2] To be performed only when Level C packing is specified. See 5.2.2, 6.2.1.

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4.10.4 Open circuit tests. Close the three pole switch. Place the ac voltmeter specified in 4.5.6 between each of the poles on the switch. Vary the three phase supply voltage so that the average input phase voltage to the high voltage power supply is 814 ± 5 volt rms.

4.10.4.1 DC voltages. (M102). Using the dc voltmeter specified in 4.5.8, measure the open circuit output voltage of circuit 1 and circuit 2 to determine conformance to 3.15.4.1.

4.10.4.2 AC voltage content. (M103). Using the bandpass filter specified in 4.5.15 with the voltmeter specified in 4.5.7 connected across the filter's terminating or output impedance, measure the open circuit ac voltage from circuit 1 and circuit 2. If a simulated generator impedance was used for this test, apply the factor necessary to convert the above ac voltage readings to the actual ac voltage content of each circuit tested. The total ac voltage content shall be measured as described above for each circuit to determine conformance to 3.15.4.2.

4.10.5 Safety check. (C1). Perform the setup specified in 4.10.4. After the open circuit output voltages reach the values measured in 4.10.4.1, remove the dc voltmeter and open the three pole switch. At 10 ± 1 minutes after the switch is opened, the open circuit output voltages of the high voltage power supply shall be measured to determine conformance to 3.15.7.

4.10.6 Load cycle 1 tests. (M104). Using the setup specified in 4.10.4, set the timer for load cycle 1 and start the repeat-cycle timer motor. Set the input voltage so that the average voltage between each line and neutral is 746 ± 5 volt rms. Connect the calibrated oscilloscope ground terminal to the common point between the dummy loads. Using the calibrated oscilloscope, measure the voltage applied to each dummy load at the instant the load relay completes the load circuits. The peak voltages and load voltage decay shall be measured to determine conformance to 3.15.5.3.

4.10.7 Load cycle 2 tests. (M105). Set the timer for load cycle 2 and repeat the test specified in 4.10.6. The peak voltages and load voltage decay shall be measured to determine conformance to 3.15.5.4.

4.10.8 Input current. (M106). Perform the setup specified in 4.10.4, set the timer for load cycle 2, and start the repeat cycle timer motor. Remove the calibrated oscilloscope from the dummy loads. Make sure no common grounds remain between the oscilloscope and load relay coil, and the high voltage power supply. Do not touch the high voltage power supply or any part of its output circuits once the oscilloscope has been removed from the dummy loads. Connect the oscilloscope across the 10 ohm resistor which is in series with the A terminal on the high voltage power supply connector.

Set the input voltage so that the average voltage between each line and the neutral is 814 ± 5 volts rms. Measure the maximum peak-to-peak voltage appearing across the 10 ohm resistor, and calculate the input current to determine conformance to 3.15.2.

4.10.9 Output impedance test. (M107). Open the three pole switch. Close the load relay for not less than five seconds to discharge the high voltage power supply output capacitors. Remove connector MS-3106A-24-27S from the high voltage power supply receptacle. Connect the 5-ohm and 680-ohm resistors specified in 4.5.14.2, in series with the output of the oscillator, specified in 4.5.19, and the output terminals of the high voltage power supply circuit 1. With the oscillator set at 60 ± 5 kHz, measure the voltage drop across the 5-ohm resistor and the output terminals of circuit 1. Use the voltmeter specified in 4.5.7. Repeat the above procedure with the output terminals of the high voltage power supply circuit 2. The impedance of the high voltage power supply circuit 1 and circuit 2 shall be calculated as follows:

$$\text{Output impedance in ohms} = 5\ \frac{\text{(Voltage across output terminals)}}{\text{(Voltage across 5-ohm resistor)}}$$

to determine conformance to 3.15.6.

4.10.10 Dielectric test (M108). Disconnect the power supply from all inputs and outputs and allow output capacitors to discharge. All pins of the receptacle MS-3102A-24-27P shall be connected together and a 60 Hz test voltage shall be applied between the connected pins and the chassis. The test voltage shall start at zero volts, shall increase, at a rate not exceeding

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500 volts per second, to 2925 ± 25 vrms, shall be kept at that level for not less than one minute, and the power supply observed for evidence of arcing or insulation breakdown.

4.10.11 Insulation resistance. (M109). Using the setup specified in 4.10.10, measure the insulation resistance at 500 vdc between the connected pins if receptacle MS-3102A-24-27P and the chassis, to determine conformance to 3.15.9.

4.10.12 Vibration. The high voltage power supply, mounted securely to a vibration device by means of the mounting holes, shall be subjected to vibration testing specified in 4.10.12.1 and 4.10.12.2.

4.10.12.1 Vibration; nonoperating condition. (M110). The high voltage power supply, mounted as specified in 4.10.12, shall be subjected to the vibration testing of Schedule I in each of three mutually perpendicular axes. Following the test, the power supply shall be examined for damage and to determine conformance to 3.16.1. See 6.2.1.

SCHEDULE I

Frequency Range	Amplitude	Minimum Test Duration		
		Room Temperature	-20°F	+130°F
10 - 30 Hz	0.060 ± 0.006 inch 1.52 ± 0.15 mm	60 min.	15 min.	15 min.
31 - 60 Hz	3g ± 0.3g	60 min.	15 min.	15 min.
61 - 100 Hz	3g ± 0.3g	60 min.	15 min.	15 min.

The frequency shall be continuously varied, between each of the limits given, at a uniform rate over a three to five minute period for the duration of the test; or, as an alternate method, the high voltage power supply shall be vibrated as specific frequencies differing by not more than five Hz in each of the test axes and covering the entire frequency range and amplitudes of Schedule I. The total test time shall be not less than that of Schedule I and the test times at the specific frequencies shall be equally divided.

4.10.12.2 Vibration; operating condition. (M111). The high voltage power supply, mounted as specified in 4.10.12, shall be subjected to the vibration testing of Schedule II in each of three mutually perpendicular axes. During and following the test, the high voltage power supply shall be examined for damage and to determine conformance to 3.16.2.

SCHEDULE II

Frequency Range	Amplitude	Minimum Test Duration Room Temperature
20 - 60 Hz	1g ± 0.1g	15 min.
61 - 150 Hz	1g ± 0.1g	15 min.

The frequency shall be continuously varied, between each of the limits given, at a uniform rate over a three to five minute period for the duration of the test; or, as an alternate method, the high voltage power supply shall be vibrated at specific frequencies differing by not more than five Hz in each of the test axes and covering the entire frequency range and amplitudes of Schedule II. The total test time shall be not less than that of Schedule II and the test times at the specific frequencies shall be equally divided.

4.10.13 Shock; operating condition. (M112). The high voltage power supply, mounted securely to a shock testing device by means of the mounting holes, shall be subjected to 3 shock pulses of not less than 8 ms duration and approximately one-half a sine wave. The pulses shall have a peak magnitude of 60 ± 6 g's and shall be applied in both directions along each of three

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mutually perpendicular axes. Following the test, the high voltage power supply shall be examined for damage and to determine conformance to 3.16.3.

4.10.14 Acceleration. (M113). The high voltage power supply, mounted on a testing device by means of the mounting holes, shall be subjected to 15 ± 2 g's acceleration for a period of not less than 0.2 second. The test shall be conducted along the longitudinal axis with the inertia not aft. Following the test, the high voltage power supply shall be examined for damage and to determine conformance to 3.16.4.

4.10.15 Temperature and humidity; nonoperating. (M114). The high voltage power supply shall be subjected to a temperature of 71° ± 3°C (160° ± 5°F) at 90 to 95 percent relative humidity for a period of 48 ± 5 hours. At the conclusion of this test period, and after returning to normal room temperature, the unit shall be examined for damage or corrosion. Prior to further testing, wiping or air drying may be performed. Drying shall not be accomplished by heating or baking. The power supply shall then be examined for conformance to 3.16.5.1.

4.10.16 Temperature tests.

4.10.16.1 High temperature; operating. (M115). The high voltage power supply shall be placed in a test chamber whose temperature shall be raised to 54° ± 3°C (130° ± 5°F) and maintained long enough for the unit to stabilize at this temperature. While still at the test temperature of 54°C (130°F), the high voltage power supply shall be subjected to the tests specified in 4.10.3 to 4.10.11, inclusive.

4.10.16.2 Low temperature; nonoperating. (M116). The high voltage power supply shall be placed in a test chamber whose temperature shall be lowered to -54° ± 3°C (-65° ± 5°F) and maintained at this temperature for a period of 48 hours ± 5 hours. At the conclusion of this test period, and after returning to normal room temperature, the unit shall be examined to determine conformance to 3.16.5.3.

4.10.16.3 Low temperature; operating. (M117). The high voltage power supply shall be placed in a test chamber whose temperature is lowered to -2° ± 3°C (28° ± 5°F) and maintained long enough for the unit to stabilize at this temperature. While at -2° ± 3°C (28°F), the high voltage power supply shall be subjected to the tests specified in 4.10.3 to 4.10.11, inclusive.

4.10.17 Pressure. (M118). The high voltage power supply, while in a nonoperating state, shall be placed in a vacuum chamber and the pressure reduced from room ambient to 4 psia +0, -2 psia (28 +0, -14 kPa) and allowed to return to room ambient for 180 cycles. Following the cycling, the pressure shall be reduced to 4 psia +0, -2 psia (28 +0, -14kPa), held for 30 minutes, and allowed to return to room ambient. The time required to complete the cycling and vacuum testing shall not exceed eight hours. At the conclusion of the test, and after having returned to ambient pressure, the high voltage power supply shall be examined for evidence of damage or oil leakage and to determine conformance to 3.16.6.

4.10.18 Radiation test. (M119). Operate the high voltage power supply as specified in 4.10.3. Using the test procedures specified in MIL-STD-462, the high voltage power supply shall be examined to determine conformance to 3.15.4.3.

4.10.19 Operation test time. (M120). Allow the high voltage power supply to operate open circuit for 9 hours and 50 ± 10 minutes. For the next 30 ± 5 minutes, draw a repeat "Load cycle 1" as specified in 3.15.5.1. At the beginning and near the end of the repeat cycle load test, measure the peak voltage applied to each dummy load to determine conformance to 3.15.5.3.

Allow the high voltage power supply to return to ambient temperature. Repeat the above procedure, but draw the repeat "Load cycle 2" as specified in 3.15.5.1 for 30 ± 5 minutes. The peak dummy load voltages shall be measured to determine conformance to 3.15.5.4.

4.10.20 High voltage test. (M122). Apply 4000 ± 100 VDC between pins G and E and 1700 ± 50 VDC between pins F and E of receptacle MS-3102A-24-27P. The test is to be conducted separately with positive voltage on pins G and F and negative voltage on pin E, with pin E grounded to the case to avoid damage to associated components. The duration of the voltage shall be not less than one minute. During the test, the high voltage power supply shall be examined for arcing or insulation breakdown.

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4.10.21 Operating life test. (M122). Operate the high voltage power supply as specified in 4.10.6, except for the input voltage, which shall be 780 ± 39 volts rms, 3 phase wye, for 30 ± 5 minutes. Shut off the input power. The high voltage power supply shall be allowed to cool to ambient temperature before restarting. For the next 30 ± 5 minutes duty cycle, operate the high voltage power supply as specified in 4.10.7, except that the input voltage shall be 780 ± 39 volts rms, 3 phase wye. Shut off the input power at the end of this duty cycle and allow the high voltage power supply to return to ambient temperature. Repeat the above procedure for a total of 199 ± 1 actual operating hours. At intervals of approximately 10 operating hours, check the high voltage power supply for compliance with 3.15.5.3 and 3.15.5.4.

4.10.22 Vibration; transportation (Level C only). (M123). The high voltage power supply, packaged and packed in accordance with Section 5 and the contract or purchase order, shall be vibrated on a barrier equipped platform in accordance with WR-11. Starting at 5 Hz, gradually increase frequency until pack just begins to leave the table. At this point, vibrate the pack for a period of one hour. Thereafter, the power supply shall be examined to determine conformance to 3.16.7.1.

4.10.23 Shock; transportation (Level C only). (M124). The high voltage power supply, preserved, packaged, and packed as specified in Section 5 and the contract or purchase order, shall be subjected to the appropriate shock tests of WR-11. Thereafter, the power supply shall be examined to determine conformance to 3.16.7.2.

5 PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified. See 6.2.1.

5.1.1 Cleaning. The high voltage power supply shall be cleaned in accordance with Method C-1 of MIL-P-116.

5.1.2 Preservation. None required.

5.1.3 Unit packaging. None required.

5.2 Packing. Packing shall be level A, B, or C, as specified. See 6.2.1.

5.2.1 Level A or Level B. The high voltage power supply shall be packed in accordance with LD 495868.

5.2.2 Level C. The high voltage power supply shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or Motor Freight Classification Rules, whichever is applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified, each shipping container shall be marked in accordance with Drawing 1803486.

6 NOTES

6.1 Intended use. The high voltage power supply is intended to be used in Torpedo MK 37 Mods 2 and 3.

6.2 Ordering data. Procurement documents invoking this specification should specify:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification and applicable drawings.
(b) Whether there are to be other than five preproduction samples. (3.2, 4.6.1)
(c) Packaging and packing levels. (3.16.7.1, 3.16.7.2, 4.10.1, 5.1, 5.2)
(d) The consequences of failure to pass tests for critical, major, and minor characteristics and the disposition of defective units. (4.2, 4.6.3, 4.7.3, 4.8.3)
(e) The number of units comprising an inspection lot if other than 300 items. (4.8.1.1)
(f) Additions, delections, and exceptions to this specification.
(g) That all additional samples required because of the failure of any units(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

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6.2.2 Contract data requirements. Items of deliverable data relevant to this specification will be delivered as identified and specified on DD form 1423 (Contract Data Requirements List) and incorporated into the contract or order. Such data usually includes certified inspection results. (4.6.2, 4.7.2, 4.8.2)

6.3 Technical cognizance. Technical cognizance of the high voltage power supplies covered by this specification is located at the Naval Underwater Systems Center, Code LA17, Newport, RI 02840.

6.4 Metrication. This document has been annotated parenthetically with metric equivalents of quantities not expressed in the International System of Units. The metric units are in accordance with ASTM E 380-72, as adopted by the DoD on 29 November 1972, and are included for the purpose of encouraging metrication. Parenthetical quantities are for information only, as they are usually not exact equivalents but have been rounded off within specified tolerances herein.

Custodian:
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Preparing Activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-P-19086E(OS) POWER SUPPLY, HIGH VOLTAGE (FOR TORPEDO MK 37 MODS 2 & 3)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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